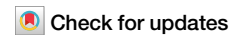


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Relatively open vegetation landscapes promoted early Pleistocene hominin evolution



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Vegetation structure and landscape openness are key ecological factors influencing human behavioural and cultural adaptation strategies. However, there is ongoing debate and lack of quantitative assessment about which vegetation landscape and openness levels were more conducive to hominin dispersal during the early Pleistocene. Here, we selected the early Pleistocene Majuangou archaeological site in China, which is the earliest site in the Nihewan Basin with reliable stratigraphic chronology and abundant archaeological materials, as the research object. We conducted pollen analysis across eight artefact layers and the natural sediments (1.75–1.29 Ma), and carried out the first quantitative reconstruction of vegetation openness. The results demonstrate that vegetation openness in the artefact layers was predominantly between 60% and 90%, while layers with vegetation openness below 50% or above 90% had either no or very few artefacts. The global comparison revealed that hominins' preference for relatively open habitats was a consistent global pattern, challenging the view that relatively closed forest vegetation landscapes were more conducive to their dispersal. Our findings suggest that enhanced resource abundance, accessibility and mobility in these environments facilitated both hominin dispersal and cultural development, highlighting the pivotal role of relatively open vegetation landscapes in shaping hominin evolution.

The relationship between vegetation ecosystems (the compositional, structural, and landscape characteristics) and the dispersal and evolution of hominins has long been a major focus in palaeoecology, archaeology, palaeoanthropology, and related disciplines^{1,2}. The early Pleistocene was a critical period for global climatic and environmental changes, characterised by cooling and aridification trends, as well as for the migration and dispersal of *Homo erectus*^{3–6}. Hominin dispersal first occurred from Africa to Eurasia^{7–9}. While existing studies have predominantly emphasised the climatic drivers of hominin evolution^{10–13}, emerging evidence underscores the crucial role of vegetation ecosystems in shaping dispersal corridors and adaptation strategies^{14–18}. The open savanna vegetation of Africa is widely regarded as having facilitated the evolution and dispersal of *H. erectus*^{19,20}. Nevertheless, the climatic patterns and vegetation types of Eurasia diverge markedly from those of Africa, rendering the African model difficult to apply to Eurasia²¹. Specifically, Eurasia may exhibit

distinct patterns of vegetation landscapes and hominin dispersal evolution.

Over the past decades, a series of innovative studies have been conducted on the relationship between hominin evolution and vegetation landscapes in Eurasia^{2,17,21}. This has resulted in two different hypotheses: (i) the open habitat hypothesis, in which hominins were better adapted to occupying open steppe or forest-steppe ecological environments^{21,22}, where the open vegetation landscape provided favourable conditions for the migration, hunting, and survival of early *H. erectus*¹⁷; (ii) the relatively closed habitat hypothesis, in which the environmental characteristics during the period when *H. erectus* was active were forest vegetation landscapes^{23,24}. This forest vegetation not only provided rich food sources for hominins but also offered safe tree-dwelling habitats^{23,25}. However, there have been no detailed studies on to which the vegetation landscape was open to be suitable for the migration, hunting and survival of early hominins.

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The main reasons for this debate and the lack of in-depth research are: (i) Issues, including discontinuous sedimentation or the poor preservation of sedimentary strata, have resulted in limited information about the changes in past vegetation ecosystems near archaeological sites^{16,26}. (ii) Most studies are limited to a single artefact layer or a relatively short time span, and have a low temporal resolution, making it difficult to comprehensively reconstruct vegetation ecosystem processes before and after the activities of *H. erectus*^{27,28}. (iii) Most research is based on qualitative analysis, and there is a lack of systematic quantitative studies of the impact of vegetation ecology and landscape characteristics on the evolution of *H. erectus*²⁹. Therefore, to better understand the influence of early Pleistocene vegetation composition and landscape openness on the migration and evolution of early hominins, it is necessary to conduct quantitative research on paleovegetation and landscape openness, based on stratigraphic sequences with continuous sedimentation, and employing multiple artefact layers, sensitive proxy indicators, a high temporal resolution, and covering a long interval.

Pollen analysis plays an irreplaceable role in revealing the paleovegetation and past landscapes openness^{30,31}. Early studies relied on non-arboreal pollen (NAP) percentages to infer vegetation openness³², but this approach fails to account for pollen productivity variations and dispersal biases, resulting in vegetation–pollen mismatches³³. Subsequent semi-quantitative alternatives, such as the arboreal/non-arboreal pollen ratio (AP/NAP)³⁴ and the difference between the maximum score of forest biomes and that of open biomes within the biome classification, have yielded improved insights but lacked precise quantification³⁵. Application of the Regional Estimates of Vegetation Abundance from Large Sites (REVEALS) model has resulted in significant research progress by incorporating pollen productivity corrections and nonlinear vegetation–pollen relationships to quantify openness, its exclusion of the bare land proportion has remained a constraint³³. To address these limitations, our study employs a transfer function methodology to establish pollen assemblage–vegetation openness relationships. This approach enhances the reconstruction reliability by utilising entire pollen spectra rather than individual taxa to minimise single-type interpretation biases³⁶.

The Nihewan Basin in northern China has the largest and most concentrated group of early Pleistocene Palaeolithic sites in East Asia³⁷. More than 280 Palaeolithic sites have been found in this region, of which 23 are older than 1.0 Ma^{38,39} (Fig. 1a, b, Supplementary Table 1). Among them, the Majuangou (MJG) archaeological site is the earliest hominin site, and it is also the earliest stone-flake tool site in Northeast Asia in terms of reliable stratigraphic relationships and abundant discovered materials⁴⁰. Ten artefact layers have been identified at the MJG site, which provide a framework for the cultural sequence of hominins from 1.76 to 1.32 Ma, in the early Pleistocene³⁹. Notably, the activity surface of a mammoth butchery site was revealed within the MJG–III artefact layer (~1.66 Ma), and more than 60 mammoth footprints were identified in MJG–II (~1.64 Ma) (Supplementary Fig. 1)³⁹. This is the earliest and the richest artefact assemblage among the early Pleistocene sites in the Nihewan Basin⁴⁰. In addition to archaeological research, environmental change research has also been conducted at the MJG site³⁷. Qualitative studies have been conducted on several artefact layers, including of mineral composition, ostracods, and animal fossils, which have revealed the burial environment and fossil animal assemblages of the artefact layers^{37,41,42}. However, quantitative studies are lacking and several key questions have not been addressed, including: What was the relationship between hominin activities and vegetation landscape openness? What degree of vegetation openness was conducive to hominin activities?

In this study, based on the existing paleomagnetic chronology and ²⁶Al/¹⁰Be burial dating^{37,43}, ages of the eight artefact layers in the section of the MJG site (40°13′31.066″N, 114°39′50.493″E, 832.6 m a.s.l.) were calibrated using the sedimentation rate (Fig. 1c). We conducted pollen analysis on 422 samples collected at 10-cm intervals from eight artefact layers and natural sedimentary layers. The REVEALS model³³ and transfer function methods³⁶ were then utilised to quantitatively reconstruct the changes in vegetation cover and landscape openness during and around the period of hominin activity in the Nihewan Basin. This study aims to reveal which types of

vegetation landscape and openness degree in the early Pleistocene of the Nihewan Basin were more conducive to attracting hominin activities. Overall, we hope to provide detailed and reliable evidence for the study of the impact of paleoenvironmental changes on the migration and evolution of hominins during the early Pleistocene in China and even the world.

Results and discussion

Stratigraphy and chronology of the MJG archaeological site

The MJG archaeological site is located in the northeastern Nihewan Basin, northern China^{40,44} (Fig. 1a, b). Due to the submersion of the bottom two artefact layers by groundwater during sampling, we selected eight artefact layers and their surrounding natural layers for analysis⁴⁰. Sedimentological analysis, incorporating grain size metrics and field stratigraphic evidence, indicates that the studied profile consists predominantly of silt (average of 70%), which is consistent with the characteristic depositional of the Nihewan Formation^{38,45} (Fig. 2, Supplementary Note 2).

The lithology of the MJG profile from bottom to top is as follows (Fig. 2): 60.5–58.1 m, grey and greyish-brown clayey silt with horizontal bedding; 58.1–42.5 m, grey and greyish-yellow clayey silt; 42.5–18 m, greyish-brown and grey silty clay with horizontal bedding (Supplementary Note 1). Samples of the MJG profile were collected continuously from the bottom upwards at 10-cm intervals, with some samples taken at 12-cm intervals. The total sampled interval was 60.5–18 m, with a thickness of ~42.5 m. A total of 422 samples were collected.

The chronology of the MJG profile was well constrained by magnetostratigraphy³⁷ and ²⁶Al/¹⁰Be burial dating⁴³. In 2004, Zhu et al.³⁷ established the magnetostratigraphic framework for the profile (Fig. 2b), confirming that the BS, MJG–I, MJG–II, and MJG–III artefact layers reside in the Matuyama reverse chron, bracketed by the Olduvai and Jaramillo normal subchrons³⁷. Based on average sedimentation rates, their ages were derived as: BS (1.32 Ma), MJG–I (1.55 Ma), MJG–II (1.64 Ma), and MJG–III (1.66 Ma)³⁷. In 2008, extrapolated ages for MJG–IV (1.69 Ma) and MJG–V (1.74 Ma) were derived using these chronologies and sedimentation rates (Fig. 2a)⁴⁴. In 2024, Tu et al.⁴³ applied ²⁶Al/¹⁰Be burial dating to BS, MJG–I, MJG–II, and MJG–III layers, with results concordant with Zhu et al.'s magnetostratigraphic ages, providing robust cross-validated chronological constraints.

Two additional artefact layers (MJG–Ia and MJG–Ib) discovered in 2013 occur stratigraphically above MJG–I⁴⁰. The MJG–Ia layer is located ~4.5 m above the MJG–I, while MJG–Ib layer is ~4.2 m above MJG–Ia and ~16.8 m below BS (Fig. 2, Supplementary Note 1 and Note 2)⁴⁰. Their ages were determined through grain size variations (Fig. 2a): The average sedimentation rate (~11 cm/ka) between MJG–I and BS was calculated from stratigraphic separation and age difference. Sediments between MJG–I and MJG–Ia show coarser grain size, while MJG–Ia to MJG–Ib intervals are finer, with a grain size difference factor of ~1.35. Based on these parameters and sedimentation rates, MJG–Ia and MJG–Ib were dated to 1.51 Ma and 1.45 Ma, respectively.

Quantification of modern vegetation landscape openness

Vegetation landscape openness, a key component of landscape ecology, is utilised in modern vegetation research to gauge the cover of non-arboreal vegetation (e.g., grasses and shrubs). This index ranges from 0 (representing 100% arboreal canopy closure, as in a dense forest) to 100% (indicating the complete absence of woody vegetation, characteristic of open grassland). Empirical studies demonstrate an inverse relationship between arboreal cover and openness: lower arboreal plant coverage rates correspond to higher vegetation landscape openness values⁴⁶.

In this study, we first assessed the consistency between the reconstructed and modern observed vegetation landscape openness. To achieve this, we conducted field validation across 57 systematically selected sites spanning a gradient from closed forest to open grassland within the study region (see Methods, Supplementary Note 4). The vegetation landscape openness quantification utilised MOD 44B Version 6 Vegetation Continuous Fields product (250 m spatial resolution), accessed via NASA's Earthdata platform

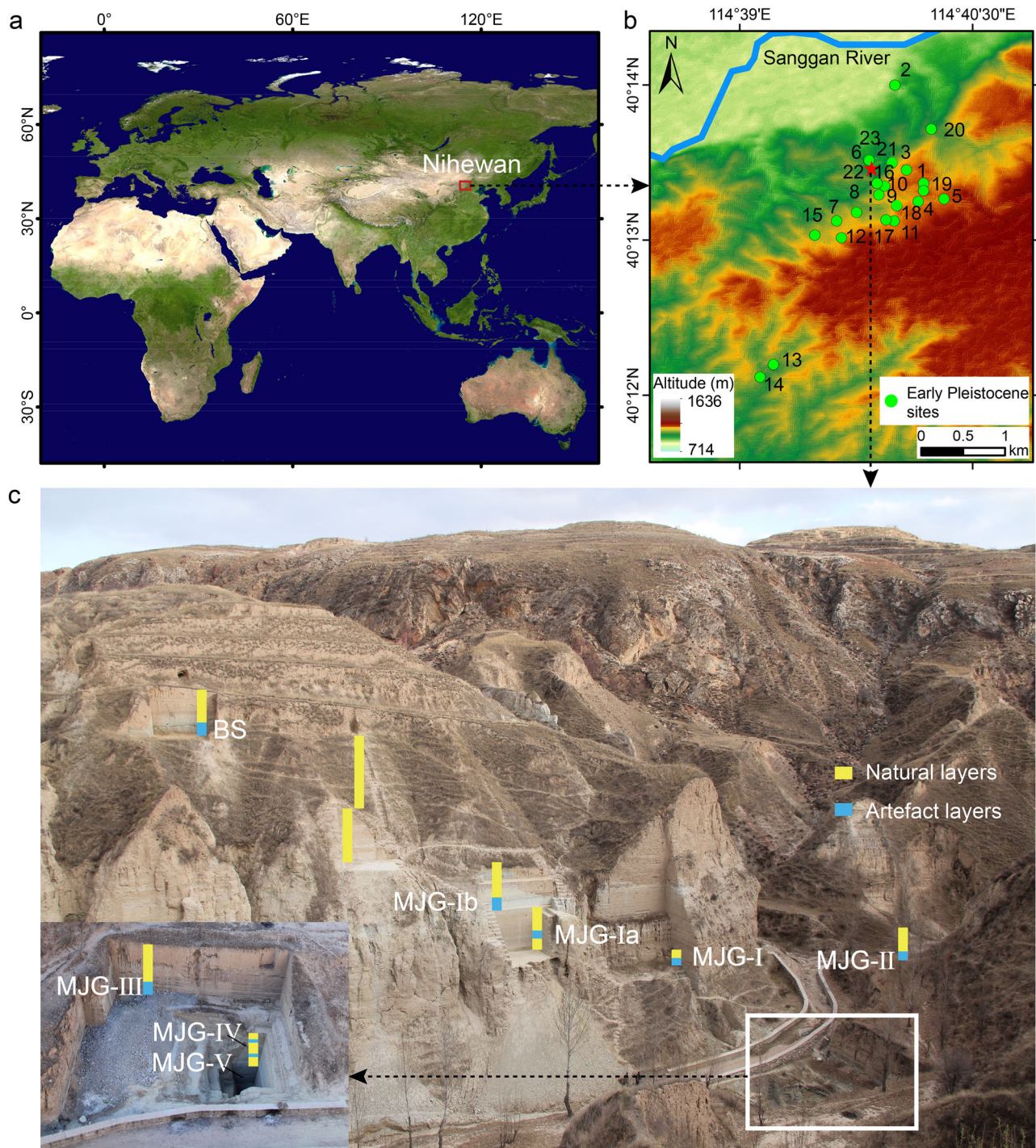


Fig. 1 | Location and field photos of the MJG archaeological site in the Nihewan Basin, China. **a** Location of the Nihewan Basin (map source: <https://search.earthdata.nasa.gov/>) (Supplementary Note 1). **b** Distribution of sites over 1 Ma in age in the Nihewan Basin. For detailed information about sites 1–23, see

Supplementary Table 1 (map source: <https://www.gscloud.cn/>). **c** Site profile and artefact layers of the MJG site in the Nihewan Basin, from Northwest to Southeast (image by Fagang Wang).

(<https://search.earthdata.nasa.gov/>)⁴⁷. This dataset quantifies three land-cover components: % tree cover, % non-tree cover, and % non-vegetated cover. For this analysis, vegetation landscape openness was calculated as:

$$\text{Vegetation landscape openness} = 100\% - \% \text{ Tree cover}$$

Application of this model showed that 85.9% of the reconstructed vegetation landscape openness samples were within the error range (Fig. 3a). The change trend between the reconstructed and observed values was also relatively consistent, with the correlation coefficient (R^2) of 0.87 ($p < 0.05$) (Fig. 3b). The reconstructed vegetation landscape openness was consistent

with the observed values, indicating that the model was dependable. This showed that this technique could then be applied to the stratigraphic data to reconstruct temporal changes in the openness of the vegetation landscape at the MJG site during the early Pleistocene.

Vegetation cover and landscape openness of the MJG site

Prior to quantitatively reconstructing vegetation landscape openness at the MJG site, we compared the performance of different distances, models, and methods (Supplementary Table 2). The comparative analysis revealed that a

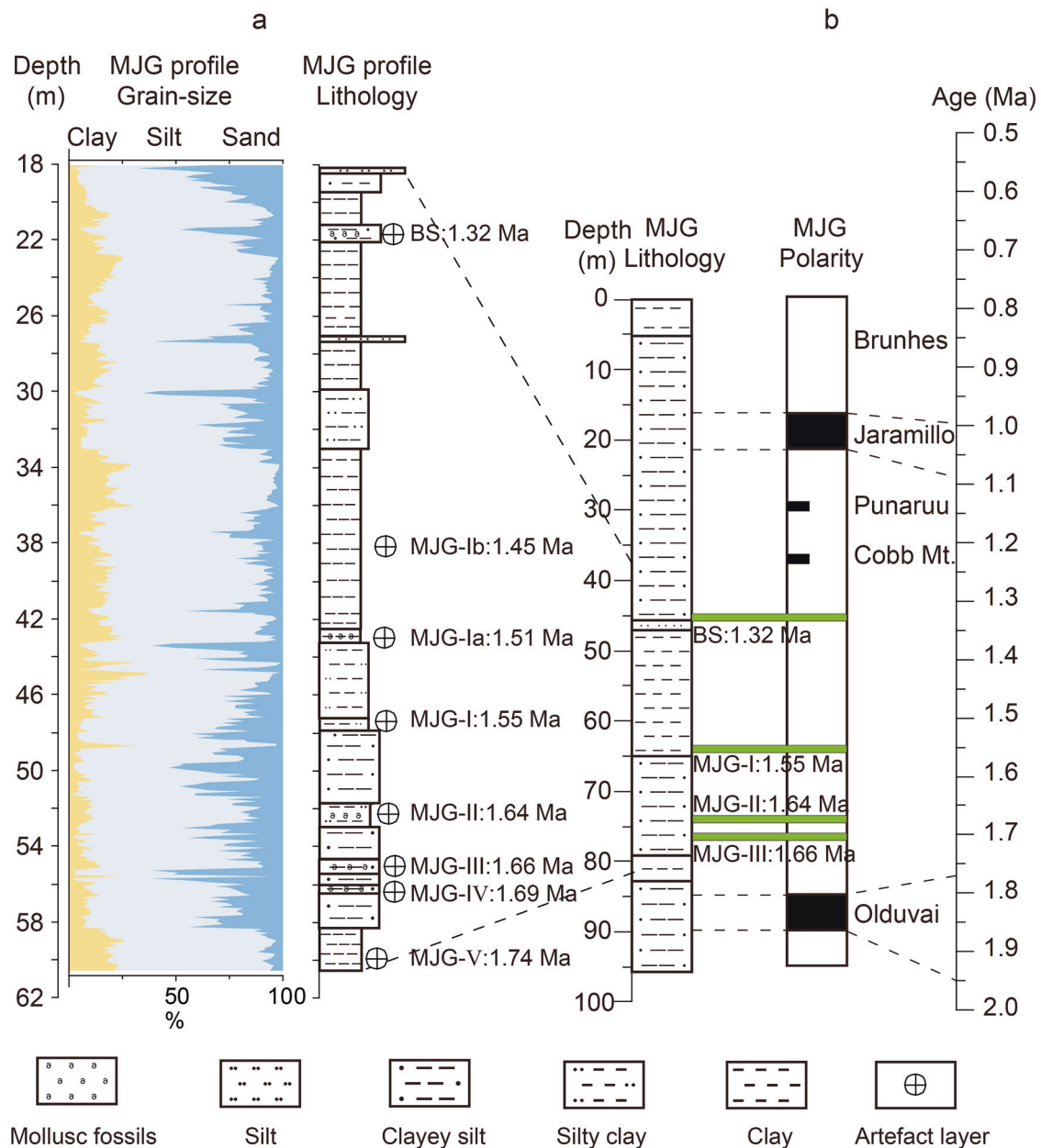


Fig. 2 | Lithology and chronology of the sedimentary profile at the MJG archaeological site. a Lithology and grain size composition of the MJG profile (Supplementary Note 2, Supplementary Data 1). **b** Lithology and chronology of the MJG profile³⁷.

1000-km radius modern pollen dataset achieved the highest predictive accuracy ($R^2 = 0.82$, Fig. S3), which was incorporated into the quantitative reconstruction of vegetation openness at the MJG site. Subsequently, we tested the degree of matching and its statistical significance between the fossil pollen assemblage of the MJG profile and modern pollen assemblage (see Methods, Supplementary Fig. 4). The MJG profile fossil pollen samples exhibit a high degree of correspondence with the regional modern pollen assemblage (Supplementary Fig. 4a), and the results of vegetation landscape openness reconstructions have statistical significance ($p < 0.05$) (Supplementary Fig. 4b). The vegetation landscape openness reconstruction outcomes are consistent with independent vegetation cover estimates (excluding bare ground) and the biome type derived from pollen assemblages (Fig. 4, Supplementary Fig. 5b, Supplementary Note 5; Supplementary Tables 3 and 4). These comparisons confirm the reliability of our quantitative reconstruction of vegetation landscape openness and vegetation cover. This result is supported by independent global climate records, including the LR04 benthic $\delta^{18}\text{O}$ stack⁴⁸ and the magnetic susceptibility

record of the Lingtai loess section in China⁴⁹ (Supplementary Fig. 5e, f). This consistency with independent multi-proxy records strengthens our confidence in the accuracy of our vegetation cover and landscape openness reconstruction.

The vegetation landscape openness of the eight artefact layers of the MJG site during the early Pleistocene is mainly between 60% and 90% (Fig. 4). The average vegetation landscape openness for each artefact layers were as follows: MJG-V (average of 88.3% and the same below), MJG-IV (66.4%), MJG-III (78.7%), MJG-II (58.5%), MJG-I (91.9%), MJG-Ia (78%), MJG-Ib (65.9%), and BS (60.8%).

Based on vegetation composition, the eight artefact layers could be divided into two main types (Fig. 4). The first group consists of six artefact layers (MJG-V, MJG-IV, MJG-III, MJG-I, MJG-Ia, and MJG-Ib), dominated by *Artemisia*, *Chenopodiaceae*, and *Poaceae*, and the herb pollen content is $>50\%$, while the tree pollen content is relatively low, $<30\%$. This suggests a forest-steppe vegetation landscape in the study area. The openness of these six layers ($>65\%$, up to 90%) further supports their

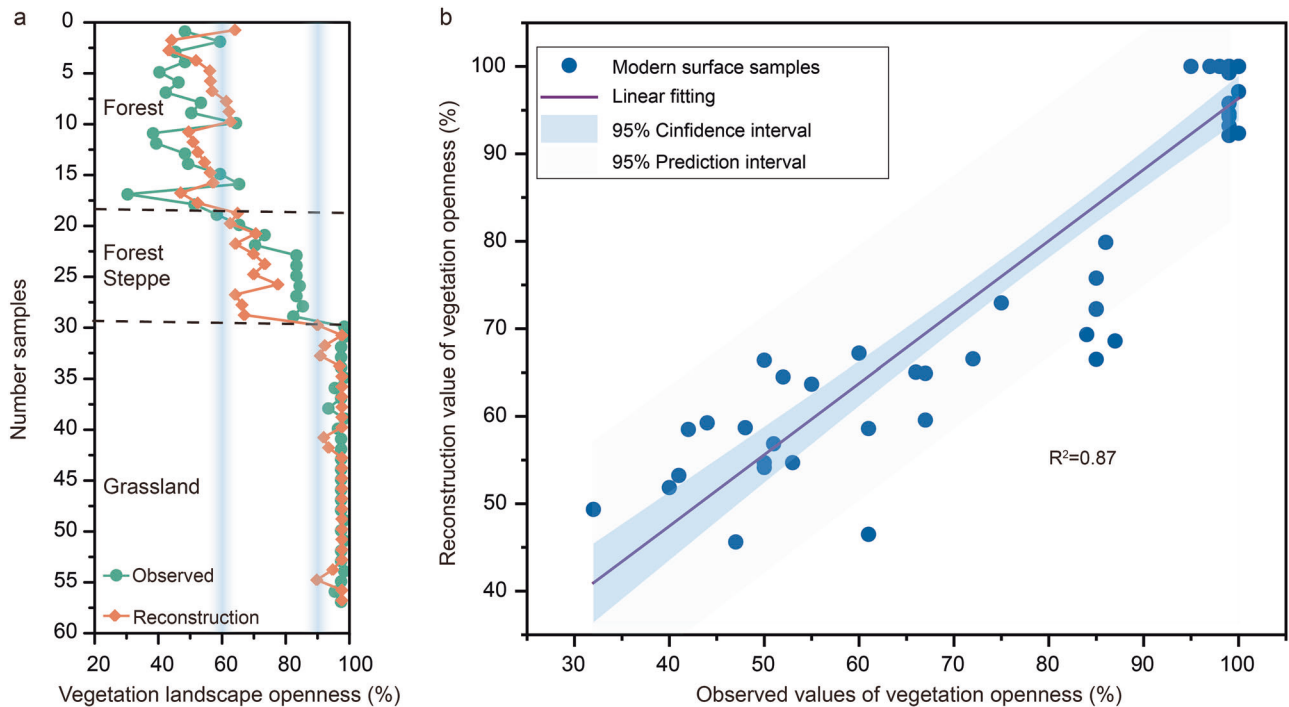


Fig. 3 | Comparison of observed and reconstructed vegetation landscape openness in modern surface samples. **a** Sample numbers 1–18 correspond to forest vegetation, 19–29 to forest-steppe, and 30–57 to grassland (see Methods, Supplementary Data 2). The figure shows observed and reconstructed vegetation landscape

openness values across three vegetation types: forest, forest-steppe, and grassland. The two light blue columns represent the critical errors for forest/forest-steppe and forest-steppe/grassland. **b** Correlation coefficient between observed and reconstructed openness values for forest, forest-steppe, and grassland vegetation landscape openness.

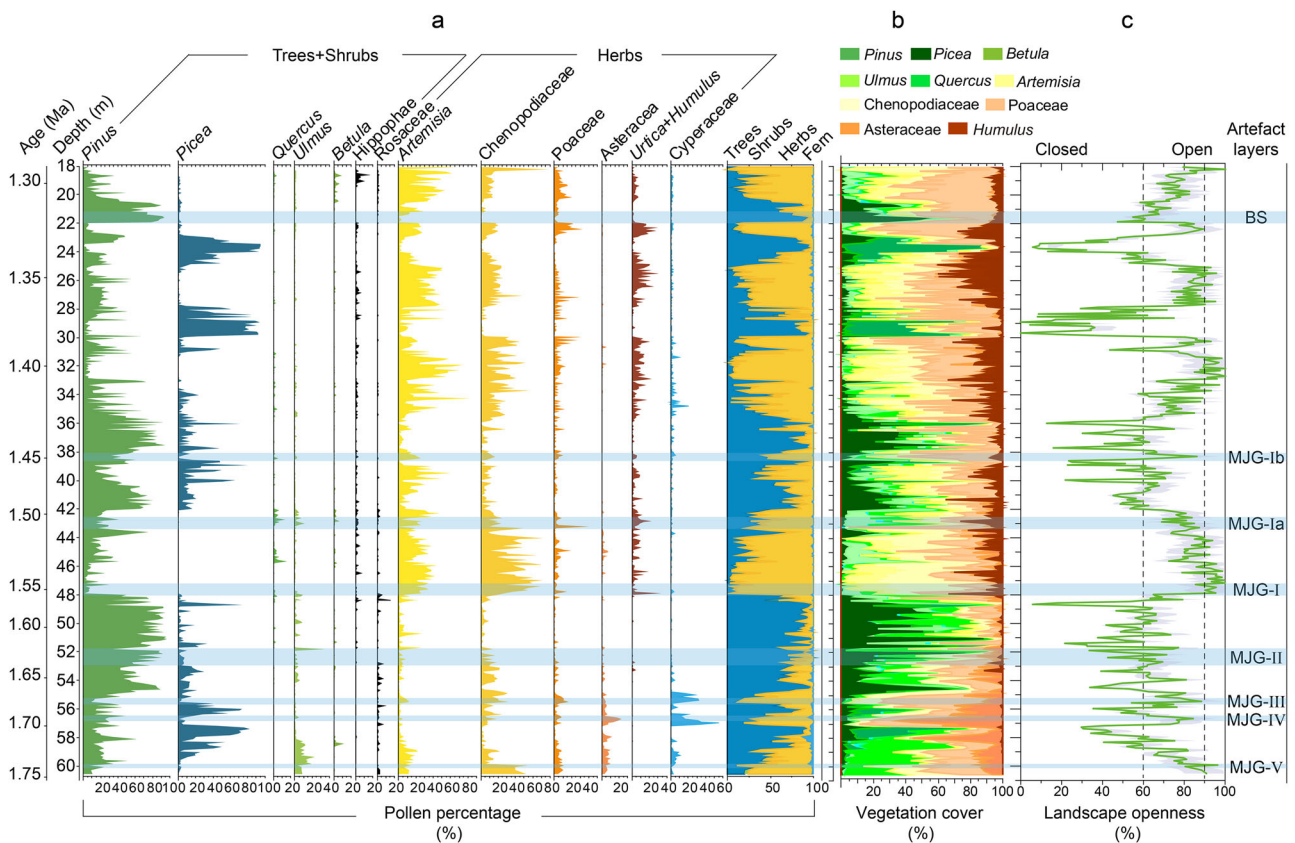


Fig. 4 | Changes in vegetation cover and landscape openness during the early Pleistocene at the MJG site. **a** Percentages of the principal pollen types. **b** Vegetation cover reconstructed based on the REVEALS model. **c** The openness of the vegetation

landscape was quantitatively reconstructed based on the transfer function method (Supplementary Note 4, Supplementary data 3).

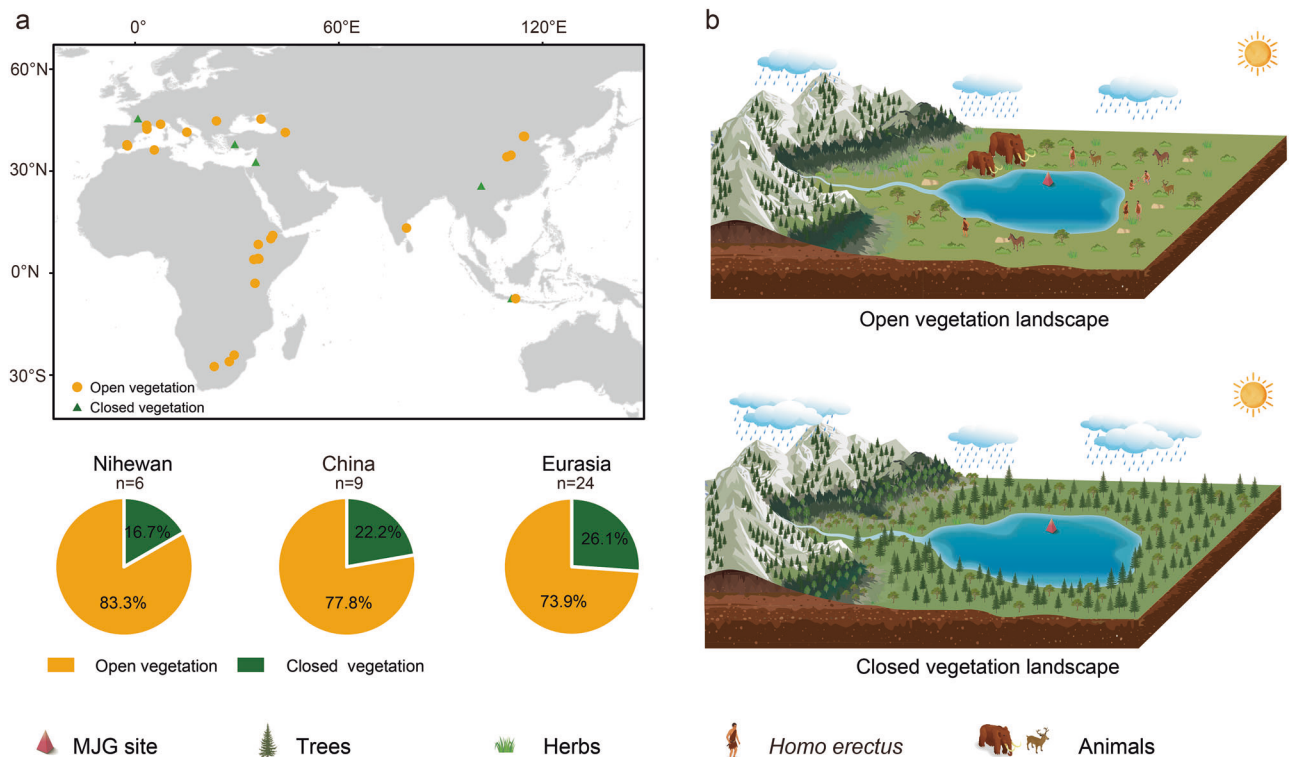


Fig. 5 | Location and vegetation landscape of global archaeological sites for the interval of 1.8–1.0 Ma. a Global distribution of 1.8–1.0 Ma archaeological sites and the proportion of open and closed vegetation landscapes in Eurasia (archaeological sites are listed in Supplementary data 4). **b** Pattern diagram of open and closed

vegetation landscapes. Credit: elements such as trees, herbs, hominins, and animals are from Huaban.com. Drawing on both these elements and the insights from this study, I reconceived and drew this diagram.

classification as forest-steppe (Fig. 3). The second group comprises two artefact layers (MJG-II, and BS), dominated by *Pinus*, with the tree pollen content up to 80%, while the herb pollen content was ~20% on average. The vegetation landscape openness at this was ~60%, which indicates an open temperate forest vegetation landscape.

However, when vegetation openness was <50% or >90%, there was minimal or no evidence of hominin activities. Where *Picea* pollen percentages exceeded 30% (e.g., in the natural layers located at the upper part of MJG-V, MJG-IV, MJG-II, and MJG-Ib) and vegetation openness was <50% (even <20% in some layers), it indicates that the vegetation landscape was a cold temperate coniferous forest. When herbaceous plants consistently accounted for a high proportion (>90%), accompanied by vegetation openness exceeding 90% (e.g., in the natural layers located at depths of 47.2–43.8 m (~1.55–1.52 Ma), 33.8–31.3 m (~1.41–1.39 Ma), and 27.7–24.8 m (~1.36–1.34 Ma)), indicating a steppe environment. In these layers, neither the steppe environments characterised by persistently excessive vegetation openness nor the forest environments with relatively high canopy density constituted the most favourable habitats for hominin.

In summary, through the comparison between the natural layers and the artefact layers, openness of the vegetation landscape reconstructed for the eight artefact layers is mainly between 60% and 90%, which indicates that a relatively open vegetation landscape was more conducive to the migration, hunting, and survival of hominins in the Nihewan Basin.

Relatively open vegetation landscape in the early Pleistocene promoted the evolution of early hominins

This study provides compelling evidence for hominin adaptation to a relatively open vegetation landscape. The openness of the vegetation landscape reconstructed from the eight artefact layers was mainly between 60% and 90%. In contrast, when the openness of the vegetation landscape was <50% or >90%, there was no or very little evidence of hominin activities. This comparison indicates that hominins selectively occupied the Nihewan

Basin during their dispersal process, where relatively open vegetation landscapes facilitated their dispersal.

To investigate whether this linkage between a relatively open vegetation landscape and hominin activity extended beyond the Nihewan Basin, we compiled all the published data from other early Pleistocene archaeological sites in Eurasia. A total of 24 archaeological sites from the early Pleistocene period (1.8–1.0 Ma) across Eurasia were collected in our study (Supplementary Data 4). Although the majority of these prior studies provided qualitative methods, we performed a statistical analysis of the available data, using the vegetation descriptions provided by the original authors (e.g., forest, forest-steppe, and grassland)⁵⁰ (Fig. 5a). Forest vegetation landscapes are categorised as closed vegetation landscapes, while forest-steppe and grassland are categorised as open vegetation landscapes. Completely open grassland vegetation landscapes, defined as those lacking any tree cover, were excluded from the final statistical comparison.

The results show that >83.3% of Nihewan archaeological sites, 77.8% of China's archaeological sites, and 73.9% of Eurasian archaeological sites had a dominantly relatively open vegetation landscape during the period of early Pleistocene hominin activity^{51–53} (Supplementary Fig. 5a). A comprehensive study also supports the inference that a relatively open vegetation landscape in the early Pleistocene was conducive to hominin evolution and migration²⁶. This common pan-Eurasian pattern suggests that a relatively open vegetation landscape was critical for the dispersal and adaptive success of *H. erectus* (Fig. 5b).

Interestingly, this preference for relatively open vegetation landscapes was not unique to Eurasia^{1,54,55}. Evidence from Africa also indicates a similar environmental preference among hominins. In northern Kenya's Turkana Basin, carbon isotope analysis ($\delta^{13}\text{C}_{\text{VPDB}}$) of pedogenic carbonates from the early Pleistocene suggests that hominins inhabited savanna ecosystems with ~40% woody cover and >60% open vegetation⁵⁴. In the Awash and Omo-Turkana Basins of eastern Africa, using stable carbon isotopes analyses from the late Pliocene epoch, combined with $^{13}\text{C}/^{12}\text{C}$ ratios from 1300 palaeosols

in adjacent areas, further indicate that woody cover was at most sites below ~40%, reinforcing the prevalence of open landscapes in hominin habitats. In east and south African, a comparative analysis of Plio-Pleistocene mammalian fossil assemblages, alongside 31 extant mammalian communities from eight different habitat types, reveals that *Homo* was the first hominid to occupy fairly open, arid grasslands⁵⁵. Across Africa, environmental data from 22 early Pleistocene archaeological sites consistently depict a predominantly open tropical savanna landscape as the primary background for early hominin evolution (Supplementary Data 4).

Conversely, relatively closed forest vegetated areas might have impeded the migration⁵⁶ and visibility of *H. erectus*, while long-term open grassland vegetation landscape areas could have been unfavourable for hominin survival due to arid climate, scarce food resources, and lack of habitat⁵⁷. This ecological constraint may explain the relative scarcity of early Pleistocene (1.8–1.0 Ma) archaeological sites in both low-latitude dense forest regions and mid-to-high latitude arid grassland zones (Fig. 5). It was precisely these vegetation characteristics, either excessively closed or overly open, that might have restricted hominin dispersal.

In summary, our findings, which are corroborated by evidence from multiple regions worldwide, indicate that relatively open vegetative landscapes were key to facilitating the migration and dispersal of early Pleistocene hominins. This pattern appears to be a prevalent and global phenomenon, deeply intertwined with the ecological preferences and adaptive strategies of *H. erectus*.

Mechanisms of hominin evolution in a relatively open vegetation landscape

During the early Pleistocene, relatively open vegetation landscapes likely played a significant role in facilitating the early hominin evolution^{17,52}. These landscapes provided an increased abundance of animal and plant resources, enhanced visibility for resource acquisition, and reduced the risk of predation, all of which favoured ecological adaptation and cultural innovations¹⁷. Our study demonstrated that such relatively open vegetation landscapes exhibited a high Simpson's diversity index (generally >0.6) and species richness, with a diverse vegetation composition, which provided diverse and abundant plant resources for early hominins (Supplementary Fig. 5c, d, Supplementary Note 3).

Moreover, sedimentological evidence indicates that the hominin occupation layers are associated with coarse-grained, lakeside facies deposits⁵⁸ (Fig. 2), which would have ensured sustained access to water resources for early hominins^{44,53}. These lakeshore environments not only promoted the growth of herbaceous plants, creating relatively open vegetation landscapes that attracted herbivores intensively exploited by hominins for hunting^{14,59}, but also provided an abundance of local lithic raw materials^{1,38,60,61}. The availability of these resources enhanced hominins' ability to exploit both animal and plant resources, thereby promoting the development of stone tool technology, as evidenced by the increased core exploitation efficiency within the MJG–Ib artefact layer⁴⁰.

The animal and plant fossils discovered in the artefact layers at the MJG site support our conclusions in this study³⁸. Specifically, fossils of steppe animals including rhinoceros, deer, and horse have been discovered in the MJG–V, MJG–IV MJG–Ia, and MJG–Ib artefact layers^{40,62} (Supplementary Note 1). Abundant mollusk shells (including *Gyraulus chihliensis* and *Planorbis youngi*), and the leaves and seeds of aquatic plants were found in MJG–III, suggesting a lakeside or swamp environment suitable for large herbivores and carnivores³⁸. A substantial quantity of fossils of *Mammuthus trogontherii*, exhibiting signs of smashing and scraping, were discovered in MJG–III (Supplementary Fig. 1). These fossils indicate scenes of hominins dismembering animals, scraping bones for meat consumption, and utilising bones, and they confirm that large animals were a crucial food source for these hominins³⁸ (Supplementary Fig. 1). The analysis of mammal fossils from the MJG–III artefact layer by Qiu et al.⁶² revealed that steppe animals accounted for ~70%, while forest animals accounted for ~15%. More than 60 intact elephant footprints were uncovered in the MJG–II artefact layer

(Supplementary Fig. 1). The analysis of biological fossils recovered from the BS layer revealed elephantids, cervids, and rhinocerotids that were adapted to a forest environment, while *Equus* was adapted to a grassland environment^{38,63}. This evidence indicates that, in addition to forest, a significant area or areas of grassland existed within the regional vegetation during the period of hominin activity⁶⁴. Qiu et al.⁶² analysed animal fossils in the BS layer and found that steppe animals accounted for ~60% of the total. Fossil evidence from the MJG site demonstrate that during the period of hominin activity, the relatively open vegetation landscape offered a diverse range of resources. The co-occurrence of fossils from both forest-adapted and grassland-adapted animals, coupled with evidence of hominin utilisation of large animals, suggests that this mixed vegetation setting was conducive to hominins survival.

Notably, similar scenarios highlighting the importance of relatively open vegetation landscapes for hominins are also evident at multiple early Pleistocene *H. erectus* archaeological sites around the world^{65–67}. In north-western Kenya in Africa, data from 481 fossil tracks, including 97 hominin footprints attributed to *H. erectus*, reveal that the open vegetation landscape of the lakeshore provided hominins with water resources, abundant food resources (including aquatic and terrestrial animals and plants), offering efficient hunting/scavenging opportunities, and enhanced visibility that facilitated migration⁶⁵. Similarly, the open vegetation landscapes in Java, Indonesia, and at Kocabaş in the Denizli Basin (Southwestern Turkey) attracted large numbers of big herbivores, providing *H. erectus* with potential animal resources for survival^{66,67}.

In summary, our results suggest that a relatively open vegetation landscape facilitated the evolution and migration of early *H. erectus*, while dense forest vegetation hindered hominin dispersal and technological development. Although climate played an important role in the migration and evolution of early hominins^{68–70}, the influence of vegetation landscape openness cannot be ignored^{2,17}. Overall, our findings offer a novel perspective on human–environment interactions during the period of hominin evolution, emphasising the role of the vegetation landscape as a key selective pressure alongside climatic change.

Methods

Pollen analysis

Pollen analysis was performed at 10-cm intervals on 422 samples, with a sample temporal resolution of approximately 1000 years. Sample preparation followed the conventional HCl–NaOH–HF treatment⁷¹. For each sample, 300 g of sediment was weighed and one tablet of *Lycopodium* spores (27,560 grains) was added to calculate the pollen concentrations. After chemical treatment, pollen and spores were extracted using heavy liquid flotation. These procedures were conducted at the School of Geographical Sciences of Hebei Normal University. Pollen identifications were made at ×400 under a Zeiss Imager A2 optical microscope with the aid of standard pollen reference publications for China⁷² and reference material preserved in the Key Laboratory of Environmental Evolution and Ecological Construction of Hebei Normal University. A minimum of 400 terrestrial pollen grains was counted for each sample. The common pollen types are illustrated in Supplementary Fig. 6. Pollen diagrams were drawn with Tilia 1.7.16⁷³.

Quantitative reconstruction of vegetation landscape openness

Sources of modern pollen and vegetation landscape openness data.

The modern pollen dataset we used is based on China's modern pollen database, comprising a total of 4164 samples⁷⁴ (Supplementary Fig. 2). The data on vegetation landscape openness corresponding to the modern sampling points are MODIS satellite data provided by the National Aeronautics and Space Administration (NASA), with a spatial resolution of 250 m (<https://search.earthdata.nasa.gov/>).

Screening the modern pollen data. Screening of modern pollen data is crucial for ensuring the accuracy of quantitative reconstructions. The selection process we applied to the modern pollen dataset employed the

following criteria: 1) Air trap and dust samples were excluded due to differences in the preservation of modern pollen compared to fossil pollen. 2) Samples with latitude, longitude, and altitude that did not match the real situation were excluded (a total of 18 samples were excluded in this study). 3) Fan et al.⁷⁵, based on modern pollen data and the Human Influence Index (HII, 0–64) (<https://sedac.ciesin.columbia.edu/>)⁷⁶, successfully distinguished the threshold values of native/secondary vegetation and secondary/anthropogenic vegetation by using the error inflection point–discriminant technique (EIPDT) (HII = 22 and 38, respectively). Vegetation begins to be affected by human activities when HII > 22; therefore, modern pollen samples with HII > 22 were deleted in this study.

In the study, the pollen assemblage was dominated by temperate taxa, with only trace amounts of subtropical and frigid zone types detected (Fig. 4). This composition suggests that the past vegetation likely developed under a temperate climatic regime. Consequently, to accurately reconstruct past vegetation landscape openness, it is crucial to select modern pollen data from a comparable spatial range. Given the dominance of temperate pollen and this regional climatic context, the space-for-time substitution approach was applied. The selection of an appropriate spatial range for the modern training set is therefore critical for model performance⁷⁷. An excessively small range may not capture the full variation within the reconstruction sequence, while an excessively large range can weaken the pollen–landscape relationship and introduce errors⁷⁷. Based on principles of quantitative reconstruction, we established modern pollen training sets within multiple distance intervals (800–1500 km from the study site, at 200-km intervals) for analysis⁷⁷. Among the tested intervals, the training set within a 1000 km radius yielded the optimal model performance (Supplementary Table 2). This optimal dataset comprised 1201 modern pollen samples (Supplementary Fig. 2). It covers a wide gradient of vegetation landscape openness (27%–100%) and includes all the principal pollen assemblage taxa found in the fossil record (Fig. 4).

Quantitative reconstruction of vegetation landscape openness. We selected the three most widely used methods for the quantitative reconstruction of vegetation openness: the Weighted Averaging Partial Least Sequence (WA-PLS), the Modern Analogue Technique (MAT)³⁶ and Random Forest (RF)⁷⁸. These three methods are currently the most widely used^{79,80}. We compared the results obtained using these three methods to select the most suitable method in this study. The predictive performances of all the calibration models were assessed via self-cross-validation, with reference to the performance statistics of each calibration model, including the root mean square error of prediction and R^2 between observed and predicted values³⁶ (Supplementary Fig. 3, Supplementary Table 2). We also used non-metric multi-dimensional scaling to test the degree of matching between modern and fossil pollen assemblages⁸¹, and the random TF method was used to evaluate the statistical significance of the pollen and landscape openness reconstructions⁸¹.

First, to verify the reliability of the reconstruction results of the chosen model, we selected 57 modern samples from the transition between forest and grassland in the vicinity of the study area (the Taihang mountains, Guancen mountains, Nihewan, and Zhangbei regions in North China), which covered the major pollen assemblages within the strata (Supplementary Note 4). Then, the vegetation landscape openness was reconstructed for two purposes: to verify the correlation between the reconstructed and observed values, and to determine the critical values of the vegetation openness of forest, forest-steppe, and grassland (Fig. 3). All these analyses were conducted using R (4.0.3) with the “rioja” (version 1.0–7) and “palaeoSigs” (version 2.0–7) packages.

Quantitative reconstruction of vegetation cover

The Landscape Reconstruction Algorithm (LRA) is based on the relative pollen productivity of the dominant plant taxa in a given study area, and the relative vegetation abundance is reconstructed using the REVEALS model³³. The REVEALS model is intended for reconstructing regional vegetation

using pollen records from large lakes (e.g., larger than 100 ha) or a combination of multiple small lakes³³. It can correct the differences in pollen productivity and dispersal characteristics among taxa, while also accounting for the type and size of sedimentary basins³³. This model has been widely applied in vegetation reconstruction from the late Pleistocene in Europe and Asia^{82–84}. The results demonstrate that, compared to unadjusted pollen percentages, the plant abundances estimated by the REVEALS model more accurately reflect true regional vegetation coverage⁸³. Previous studies have shown that the Nihewan paleo-lake began to form during the late Pliocene⁸⁵. The paleo-lake covered an area of ~10,000 km² between 2.18 and 1.87 Ma, being the largest at the time⁸⁵. From 1.87 to 0.73 Ma, it retreated by 3–4 km, reducing its area to an estimated ~8000 km²⁸⁵. Consequently, since this conforms to the underlying assumption of large lakes, the REVEALS model is appropriate for reconstructing past vegetation cover in this region.

The REVEALS model was run using the LRA. REVEALS.v6.2.2 with model parameters consisting of the radius of the depositional area (m), the fossil pollen count, pollen fall speeds (m/s), relative pollen productivity, and the regional vegetation range and its variance–covariance matrix³³. The relative pollen productivity values of 10 plant taxa were selected (Supplementary Table 5), and their pollen percentages accounted for ~90% of the pollen assemblages, covering the principal pollen types in the strata. The pollen productivity of tree taxa was obtained from data on the relative pollen productivity of mountain areas in the upper reaches of the Sanggan River in the Nihewan Basin in Guancen, in the same research area; this was because the pollen productivity of tree taxa is close in the same climate area⁸⁶. The pollen productivity of herbaceous plants is affected by a wide range of climatic factors, and therefore, the relative pollen productivity of herbaceous plants was selected based on integrated research results for northern China⁸⁷.

Data availability

All data (Supplementary Data 1–4) used in this study are freely available online (<https://doi.org/10.6084/m9.figshare.30415381>).

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Author contributions

B.F., Y.L., and H.L. conceived the study; Q.X. provided the modern pollen data; F.W. and F.X. provided field materials; B.F., Z.Z., Y.G., and B.L. collected the samples; B.F., J.Y., and Z.Z. treated and identified the pollen samples; B.F. and D.X. performed statistical analyses; B.F., Y.L., H.L., C.D., and Q.H. wrote the text. All authors commented on the interpretation of the results and gave final approval for publication.

Competing interests

The authors declare no competing interests.

Additional information

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